F18CE2 MV Calculus and Real Analysis B: Tutorial 4

1 State the Bolzano-Weierstrass theorem.

(a) Give an example of a sequence (a_n) with $6 \leqslant a_n \leqslant 8$ for all n and (a_n) not convergent. Write down a convergent subsequence of (a_n). Give an example of a bounded sequence which is not convergent, and write down a convergent subsequence of this sequence.

(b) The sequence (a_n) is given by

$$a_n = \frac{3n\cos(2n^2+1)}{n+2}\,.$$

Explain why (a_n) has a convergent subsequence (you don't need to find one!).

2 For $x > 0$ the sequence $(a_n(x))$ is given by

$$a_n(x) = n\left(x^{1/n} - 1\right).$$

It can be shown that $\lim_{n\to\infty} a_n(x) = h(x)$ exists (you do not need to prove this).

By considering $\lim_{n\to\infty} a_{2n}(x)$ show that $h(x) = 2h(\sqrt{x})$.

3 Define what is meant by $\lim_{x\to\infty} f(x) = L$. Using this definition, prove that

$$\lim_{x\to\infty} \frac{3x}{x^2+2} = 0.$$

4 Define what is meant by $\lim_{x\to\infty} f(x) = \infty$. Using this definition, prove that $\lim_{x\to\infty} (x^2 - 3x) = \infty$.

5 Define what is meant by $\lim_{x\to a} f(x) = L$. Using this definition, prove the following results.

(a) $\lim_{x\to 2} (4x-1) = 7$.

(b) $\lim_{x\to 6} x^2 = 36$.

(c) $\lim_{x\to 2} (x^3 - 3x) = 2$.

(d) $\lim_{x\to 1} \dfrac{x+2}{x+5} = \dfrac{1}{2}$.

6 Suppose that $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$. Prove directly from the definition of limit that

$$\lim_{x\to a} (2f(x) + 3g(x)) = 2L + 3M.$$

7 Suppose that $\lim_{x\to a} f(x) = L$, and that $L > 0$. Prove that there exists $\delta > 0$ such that

$$f(x) > \frac{L}{2}, \quad \text{for } 0 < |x-a| < \delta.$$